Understand the User, Don't Repeat: Linguistic Cues of Unmet Needs and Explicit Repair in Task-Oriented Dialogue

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Abstract

When a dialogue system fails, users seldom say so in emotional words. They repeat the request, rephrase it, write "what?", or get shorter. We study these cues in EmoWOZ, a corpus with emotion labels on every user turn. Human operators change their replies after a dissatisfied turn (acknowledgment rises from 3.9% to 17.8%); a deployed machine system does not (3.7% vs. 3.5%). Toward the machine, 46.8% of dissatisfied turns repeat an earlier request, and only 7.7% contain a complaint or negation word, so word-level sentiment misses most of them. We then ask whether a language model can be made to answer the unmet need rather than the last message. Asking the model to "read the user first" helps a strong model but hurts an 8B model, which misreads the need and apologizes instead of delivering. We propose REPAIR, which reads the dialogue explicitly — repair cues, a ledger of requested, delivered and pending items, and database rows for the pending item — and hands this reading to the generator. On 100 unseen dialogue cases, REPAIR raises how well an 8B model resolves the user's need from 0.45 to 1.22 on a 0–2 scale (0 = unresolved, 1 = partly, 2 = delivered), cuts the share of replies after which the user would likely give up from 49% (prompt-only reading) to 19%, and brings the 8B model to the level of a much stronger model given a "be concrete" instruction.

1 Introduction

A user who has asked twice for a guesthouse in the south does not need a third question about the price range, and does not need an apology. The user needs the guesthouse. Dialogue systems, including systems built on large language models, are usually asked to respond to the last message. When an earlier request is still open, the last message is often a repetition ("hotel", "hotel options", "four star hotels"), and a system that answers it in isolation repeats its own failure.

Conversation analysis calls these moves *other-initiated repair*: the listener signals that something was not understood or not delivered and invites the speaker to fix it [Schegloff et al., 1977]. In the terms of Clark and Brennan [1991], the request has not been grounded. Our view is that unmet needs are visible mostly in this linguistic behaviour — repetition, rephrasing, repair tokens, shortening — and only partly in emotional vocabulary. Earlier work by the author on offensive language took the same position: lexical and syntactic structure, together with a user's writing style, carried most of the signal [Chen et al., 2012].

This paper makes three contributions. (1) A corpus study on EmoWOZ [Feng et al., 2022] and DBDC3 [Higashinaka et al., 2016] showing that a machine system does not react to dissatisfaction, that users express dissatisfaction in task dialogue mainly by climbing a repair ladder (dissatisfaction rises from 16.3% at stage 0 to 79.3% at stage 3+, capturing 81.7% of negative turns vs. 7.7% for

complaint words), and that repetition is task-specific. (2) A negative result: telling a model to infer the user's state from the dialogue helps a strong model and hurts a small one. (3) REPAIR, an explicit reading of repair cues and pending needs, grounded in the task database, which makes a small model resolve the user's need far more often on unseen dialogues and outperforms ledger-only grounding on high-escalation turns. We release code, prompts and all model outputs.

2 Related Work

Emotion in task-oriented dialogue. EmoWOZ labels every user turn of MultiWOZ [Budzianowski et al., 2018] and of 1,000 human–machine dialogues (DialMAGE) with one of seven emotions [Feng et al., 2022]. Recent work detects frustration in deployed systems [Hernandez Caralt et al., 2024] and studies emotion in customer-service dialogue, finding that apology and empathy often leave anger in place [Labat et al., 2025]. These works detect the state; we study how it is expressed and how to act on it.

Repair and grounding. Our cue set follows the repair literature [Schegloff et al., 1977] and grounding theory [Clark and Brennan, 1991].

Dialogue breakdown detection. The DBDC challenge series [Higashinaka et al., 2016] annotates system turns as breakdowns in chat-oriented dialogue. We use DBDC3 as a contrast corpus: without task goals, user repetition barely rises after breakdowns.

Dialogue state tracking. Classical dialogue state trackers infer slot–value belief states across turns [Budzianowski et al., 2018]. REPAIR is a lightweight, zero-shot alternative focused on *repair and grounding state*: rather than training a neural tracker, it pairs surface repair cues with a ledger of requested versus delivered items so any off-the-shelf generator can close pending needs without schema-specific fine-tuning.

LLM judges. LLM judges have known biases [Zheng et al., 2023]; we report one more in task dialogue: they reward confident, invented bookings.

3 Corpus Study

EmoWOZ has two parts: MultiWOZ, where a human "wizard" plays the system (10,438 dialogues), and DialMAGE, where a machine system talks to people (995 dialogues). We call a user turn *negative* if it is labelled dissatisfied or abusive. Confidence intervals are 95% bootstrap intervals over dialogues (2,000 resamples).

The machine does not react. We match apology and acknowledgment words (*sorry, apologize, my mistake, I understand*, ...) in the system's next reply. Human wizards use them after 17.8% of negative turns and 3.9% of neutral turns. The machine: 3.5% and 3.7%. Including *unfortunately / I'm afraid*: wizards 20.3% [17.7, 22.7] vs. 5.4%; machine 5.0% [4.2, 5.8] vs. 5.6% [5.0, 6.3].

Dissatisfaction is expressed by repeating. Table 1 shows, for each negative turn, whether it contains a complaint or negation word, and whether it repeats an earlier request of the same user (token Jaccard ≥ 0.5).

	n	complaint word	repeats own request	neither
MultiWOZ, negative turns	958	53.0%	3.0%	44.6%
DialMAGE, negative turns	4,264	7.7%	46.8%	47.6%
DialMAGE, neutral turns	7,230	3.0%	13.2%	84.2%

Table 1: How dissatisfaction is expressed. With a human wizard users complain in words; with a machine they repeat themselves.

When the machine repeats its own previous reply (Jaccard ≥ 0.8) after a negative turn, the next user turn is still negative 88.3% [80.0, 95.7] of the time, against 54.6% [52.7, 56.6] otherwise ($n = 71$ repeats).

Is repetition specific to task dialogue, and does the finding hold on other task corpora? In the DBDC3 chat-oriented corpus [Higashinaka et al., 2016] (415 dialogues, no task goals), users repeat after a system breakdown only 8.4% [6.5, 10.4] of the time, barely above the 6.8% [5.9, 7.8] baseline (+1.5 pp). In DialMAGE the gap is +33.6 pp: users repeat when they have an unmet *task* need, not simply when the system is incoherent. To verify that our cue findings are not specific to MultiWOZ domains, we apply the same zero-shot repair reader (`analyze_uss.py`) to two external goal-oriented benchmarks annotated with turn-level user satisfaction [Sun et al., 2021]: Schema-Guided Dialogue [SGD; Rastogi et al., 2020] (1,000 dialogues across 20 domains such as banks, flights, media and doctors; 9,833 user turns, 2,194 dissatisfied) and ReDial movie recommendation (999 dialogues; 7,619 user turns, 1,793 dissatisfied). Explicit emotion words (*frustrated, annoyed, useless, angry*, ...) appear in only **0.3%** [0.1, 0.5] of dissatisfied turns in SGD and **2.7%** [1.9, 3.6] in ReDial, whereas REPAIR's surface repair cues (stage ≥ 1) capture **63.6%** [61.4, 65.8] and **51.2%** [48.9, 53.8], doubling the dissatisfaction rate over stage 0 in SGD (30.5% vs. 15.2%) and raising it by +12.3 pp in ReDial (31.1% vs. 18.8%).

These findings motivate a method that detects repetition-based repair cues and tracks pending needs, rather than relying on sentiment words or the model's own reflection.

4 REPAIR

REPAIR (Reading Escalation And Pending Items for Response) builds a short, checkable reading of the dialogue at the current user turn. It is deterministic and uses no model calls.

1. **Repair reader.** Cues in the last user turn: repetition (Jaccard ≥ 0.5 with an earlier user turn) or rephrasing (0.3–0.5); repair tokens (*what?*, *??*, *I said*, *already*); negation (*no, not, again, still*); talk about the conversation (*can you help*, *understand*); a turn under half the user's median length; give-up words. These map to an escalation stage 0–4.

2. **Grounding ledger.** For each domain (restaurant, hotel, attraction, train, taxi, ...), the constraints the user stated, whether a matching entity was delivered by the system afterwards, and which required details are missing (train: origin, destination, day; taxi: origin, destination, time). Turns without domain words refine the current domain; a bare answer fills the slot the system asked for; negated needs ("does not need wifi") remove a constraint. Only the current topic is shown.

3. **Database grounding.** Rows from the MultiWOZ database that match the pending request;

if none, constraints are relaxed one at a time (type and amenities first) and the relaxation is stated.

4. **Policy.** By stage: deliver the pending item first; ask only for missing required details, offering likely choices; book when asked. At stages 3–4 the model *re-grounds* before delivering: one line restating what it understood the user wants (Clark and Brennan's presentation phase), then the best database option and at most two alternatives. It never hands the user to a person: escalation volume is exactly what a human team cannot absorb, so recovery has to happen in the model's own next turn.

The reading is appended to the dialogue, together with the system's last turns marked "do not repeat". Any generator can use it. Among REPAIR's four components, the repair reader and the escalation policy are completely domain-agnostic (as validated zero-shot on SGD's 20 domains above); for the grounding ledger and database lookup, all entity names, categorical slot values (e.g. food types, attraction types, stations) and searchable columns are induced directly from the target JSON database tables at startup rather than hand-listed per venue, so adapting to a new schema requires only supplying its database table and a short map of domain trigger words.

The repair ladder tracks emotion across the corpus. Applying REPAIR's stage reader and grounding ledger to all 12,093 annotated user turns in DialMAGE (`analyze_ladder.py`), user dissatisfaction rises monotonically up the repair ladder (Table 2): from 16.3% at stage 0 (no repair cues) to 34.3% at stage 1 (rephrasing, negation or shortening), 60.3% at stage 2 (repeating an earlier request or other-initiated repair), and 79.3% at stage 3+ (repeated repair or meta-complaint), capturing 81.7% of all negative turns at stage ≥ 1 (vs. 7.7% for complaint words). Next-turn dissatisfaction likewise climbs from 32.7% after stage 0 to 57.1% after stage 3+. Independently, turns where the grounding ledger shows an undelivered request from an earlier turn are negative 40.5% [38.9, 42.1] of the time ($n = 7{,}919$) versus 25.3% [23.3, 27.3] otherwise ($n = 4{,}174$).

Repair ladder stage	n	current negative	next turn negative
Stage 0 (no repair cues)	4,767	16.3% [15.0, 17.7]	32.7% [31.1, 34.2]
Stage 1 (rephrase, negation, shorter)	4,242	34.3% [32.5, 36.1]	37.9% [36.0, 39.7]
Stage 2 (repeat 1×, *what?*, meta)	2,177	60.3% [58.0, 62.5]	45.3% [42.8, 47.8]
Stage 3+ (repeat $\geq 2\times$ or + OIR/meta)	907	**79.3%** [75.7, 82.7]	**57.1%** [52.8, 61.4]

Table 2: User dissatisfaction by repair escalation stage across all 12,093 DialMAGE user turns (95% bootstrap CIs over dialogues). The emotion label follows the repair ladder rather than emotional vocabulary.

5 Experiments

Cases. From DialMAGE we take every dialogue at the first user turn that is negative and repeats an earlier request: the need is still unmet. We developed prompts and the reader on 60 cases of the test split, and on 81 cases of the dev split for a second round; the results we report are on 100 cases from the train split, which were not looked at during development.

Conditions. A: reply to the last message. B: be concrete, answer rather than ask back. C: prompt-only reading — find what is unresolved from the same linguistic cues and how the user

feels, then deliver it (for the 8B model, two calls: read, then reply). D: REPAIR. E: REPAIR without stage, cues and policy (ledger and database only). To test whether REPAIR's gain is merely from database access, we also test two fair database-grounded baselines that use the exact same database, query relaxation and row formatting as REPAIR: F (last-message RAG, retrieving rows from the user's last turn alone) and G (LLM-tracked dialogue state, where the generator itself reads the full dialogue, outputs JSON `{domain, constraints}`, and queries the database).

Generators. Claude 3.5 Sonnet [Anthropic, 2024] via the `claude` CLI, and Llama 3 8B [Grattafiori et al., 2024] (dolphin-llama3:8b via Ollama). A–C have no database; D and E have the rows REPAIR selects.

Evaluation. Two judge models from different families (Claude 3 Opus and Gemini 2.5 [Gemini Team et al., 2024]) independently see the dialogue and the five replies in random order, and score each reply: *resolves* (0 does not, 1 partly, 2 delivers the need), *give_up* (a user in this state would likely give up), *repeats*, and the best reply. In a second pass each judge also sees the database rows for the current request and is told that a contradicting fact cannot score 2. Across all 500 judged replies per run, the two judge families agree within 1 point on *resolves* 96.4%–100.0% of the time; fact-checking raises inter-judge correlation from $r = 0.73$ to $r = 0.90$ on the 8B run (exact agreement 59.4% to 85.2%) and from $r = 0.66$ to $r = 0.81$ on the Sonnet run. An objective check without any model asks whether the reply names a real database entity that satisfies the pending constraints. Differences are paired bootstrap 95% intervals over cases.

5.1 Results

	Claude Opus judge		Gemini judge		
Llama 3 8B	resolves	give up	resolves	give up	real & match
A last message	0.45	29%	0.98	19%	23.0%
B be concrete	0.45	28%	0.98	18%	21.8%
C prompt-only reading	0.33	49%	0.73	35%	18.4%
D REPAIR	**1.22**	**19%**	**1.51**	**14%**	60.9%
E ledger + database	1.16	23%	1.39	22%	62.1%
A, fact-checked judge	0.20	54%	0.26	41%	
B, fact-checked judge	0.17	56%	0.26	39%	
C, fact-checked judge	0.20	69%	0.25	51%	
D, fact-checked judge	**1.41**	**14%**	**1.49**	**12%**	
E, fact-checked judge	1.31	17%	1.39	16%	

Table 3: 100 unseen DialMAGE dialogues, Llama 3 8B generator, judged independently by Claude Opus and Gemini (both different model families from the generator). "Real & match" is over the 87 cases with a database request (the original DialMAGE system achieved 10.3%, after which 53.6% of users stayed negative).

Small model. Under both judges, REPAIR resolves the need far more often than every baseline (Table 3): with Claude Opus, D−B +0.77 [0.55, 0.98] and D−C +0.89 [0.70, 1.08]; with Gemini, D−B +0.53 [0.33, 0.72] and D−C +0.78 [0.60, 0.96]. Under both judges, prompt-only reading

(C) significantly hurts the 8B model compared to B (Opus −0.12, Gemini −0.25 [−0.42, −0.08], give-ups +17 pts): inspecting 18 development cases where C lost, the model's own reading named needs from earlier in the dialogue or invented them ("Indian restaurant" for a guesthouse request), and the reply apologized and promised ("please bear with me") instead of delivering. The same pattern held on the 81 dev-split cases (D 1.10 vs. C 0.57). Across all 241 evaluated cases spanning all three DialMAGE splits (213 with a database request), REPAIR raises objective real, matching entity delivery from 18.3% (A), 20.6% (B) and 17.4% (C) to **62.9%** (D) on the 8B model, and from 29.0% (A), 41.6% (B) and 45.8% (C) to **74.3%** (D) on Claude Sonnet.

Strong model. With Claude Sonnet as the generator (Table 4), plain judges without database verification favour prompt-only reading. However, fact-checked judging reverses the picture under both judges: with Opus, D resolves 1.64 [1.52, 1.75] vs. C 0.82 and B 0.76 (D−C +0.82 [0.66, 0.97], D−B +0.88 [0.71, 1.05]); with Gemini (cross-family), D resolves **1.94** [1.88, 1.99] vs. C 1.04 [0.90, 1.19] and B 0.91 [0.77, 1.05] (D−C +0.90 [0.76, 1.04], D−B +1.03 [0.89, 1.17]; D picked best 53% vs. C 6%). REPAIR raises real, matching entities from 52.9% (C) to 67.8% and cuts entities that violate the user's constraints from 33.3% to 13.8%. Plain judges reward confident invented bookings (see below); fact-checked judges penalise them.

	Opus judge		Gemini judge		
Claude Sonnet	resolves	give up	resolves	give up	real & match
A last message	0.83	35%	1.19	21%	31.0%
B be concrete	1.25	7%	1.62	2%	43.7%
C prompt-only reading	1.45	2%	1.85	0%	52.9%
D REPAIR	1.34	18%	1.55	14%	67.8%
E ledger + database	1.28	19%	1.49	14%	65.5%
A, fact-checked judge	0.68	38%	0.89	24%	
B, fact-checked judge	0.76	19%	0.91	8%	
C, fact-checked judge	0.82	12%	1.04	0%	
D, fact-checked judge	**1.64**	**3%**	**1.94**	**1%**	
E, fact-checked judge	1.65	3%	1.90	1%	

Table 4: Same 100 cases, Claude Sonnet generator. Plain judges reward invented bookings; fact-checked judges penalise them. Gemini is a cross-family judge for both generator and first judge.

Small model with REPAIR vs. strong model. On the same 100 cases, Llama 8B with REPAIR is close to Claude Sonnet told to be concrete under both judges (Opus: 1.22 vs. 1.25; Gemini: 1.51 vs. 1.62) and above Claude answering the last message (Opus: 0.83; Gemini: 1.19).

What carries the gain. D and E do not differ significantly on the 8B model overall (D−E resolves: Opus +0.06 [−0.07, 0.19]; Gemini +0.12 [−0.05, 0.28]): on the 88 stage-2 cases the explicit ledger and the database account for the entire gain. However, on the 12 stage-3 cases — where the user shows stronger repair escalation — D significantly outperforms E on the 8B model: Opus D−E +0.50 [0.08, 0.92] (D 1.50 vs. E 1.00), Gemini +0.67 [0.25, 1.17] (D 1.92 vs. E 1.25), and give-ups drop from 33% (E) to 0–8% (D). For instance, when a user has asked three times for a 3-star hotel in the east (none exists in the database, case DMAGE271#8), E stalls with "Let

me check if there are any hotels," whereas D's stage-3 policy immediately delivers the closest real match (Express by Holiday Inn, 2 stars in the east, with address and phone), scoring 2 under both judges against 0 for E. The escalation-aware policy helps precisely where users are most frustrated.

Is the gain just database access? Fair RAG and LLM-DST baselines. Because A–C have no database, a natural question is whether any database lookup would suffice. Evaluating F (last-message RAG) and G (LLM-tracked DST) with the exact same database and query function (`runs/final_{llama,claude}_fg`) shows that database access alone falls far short of REPAIR. On the objective entity check ($n = 87$), F delivers a valid matching entity in 46.0% of cases on Llama 8B (naming a constraint-violating entity in 35.6%) and G in 42.5% (26.4% invalid), versus **62.1%** valid (**11.5%** invalid) for D; on Claude Sonnet, F achieves 50.6% valid (24.1% invalid) and G 53.8% valid (25.6% invalid), versus **67.8%** valid (**13.8%** invalid) for D. In blind 7-way evaluation (candidates T–Z judged together by Gemini), D with the re-grounding policy (restate the need in one line, deliver the best option, never hand off to a human) significantly beats both F and G: on Llama 8B ($n = 100$), plain D scores **1.44** vs. F 1.06 (D−F +0.38 [0.18, 0.58]) and G 0.97 (D−G +0.47 [0.27, 0.67], give-ups 16% vs. 38%), and fact-checked D scores **1.44** vs. F 0.85 (D−F +0.59 [0.37, 0.80]) and G 0.78 (D−G +0.66 [0.46, 0.87], give-ups 15% vs. 39%). On Claude Sonnet ($n = 90$), re-grounding eliminates the plain-judge gap — D leads under both plain judging (**1.81**, give-ups **2.2%**, vs. G 1.77, C 1.72, B 1.53 [D−B +0.28], F 1.28 [D−F +0.53]) and fact-checked judging (**1.92**, give-ups **1.1%**, vs. G 1.58 [D−G +0.34 [0.21, 0.49]] and F 1.21 [D−F +0.71 [0.52, 0.90]]). When users are frustrated, their last turn is often an elliptical repair fragment ("postcode?", "any area", "book it"), so F loses earlier constraints, while G's LLM state tracker is misled by failed system turns and earlier completed domains. Tracking *requested minus delivered* explicitly is what makes the database useful.

Judges reward invented bookings. Across all 84 Sonnet replies that named a database entity violating the user's constraints (`named_invalid`), plain judges still awarded full marks (*resolves* = 2) 38% of the time under Opus and 65% under Gemini; fact-checked judging cuts this to 17% and 21% (and on the 8B run from 16% and 44% down to 10% under both judges). For example, without the database, both judges gave full marks to a reply that "booked" train TR0899 for a Leicester–Cambridge request; in the database TR0899 runs Cambridge–London. Showing the judge the database rows removes this: for the 8B model, B falls from 0.45 to 0.17 (Opus) and from 0.98 to 0.26 (Gemini), while D is 1.41 (Opus) and 1.49 (Gemini). Because the rows come from the same reader REPAIR uses, we report both passes.

6 Limitations

Evaluation uses two LLM judge families (Claude Opus and Gemini) plus an objective entity check; we do not include a separate human rating study on the generated replies, relying instead on EmoWOZ's human emotion annotations, DBDC3's 30-annotator breakdown labels, and exact database matching. The complaint-word list is lexical: on 200 sampled negative turns it matched 7.0%; a simple rule flags 1 of the 14 matches as task language ("no, I need a train..."), but we have not yet hand-labelled the matches, so we report no precision figure. The low match rate supports our claim that word-level sentiment misses most dissatisfaction. The repeat threshold (Jaccard ≥ 0.5, yielding 46.8%) is not sensitive to the exact value: 0.4 gives 53.7%, 0.6 gives 37.0%. On 20 sampled cases the REPAIR reader named a pending domain in 19 and returned database rows in 13; whether those readings are correct has not yet been checked against hand labels (sheets in `validation/`).

Our corpus findings replicate across four datasets (EmoWOZ-MultiWOZ, EmoWOZ-DialMAGE, USS-SGD across 20 domains, and USS-ReDial), with DBDC3 as a chat-oriented contrast; the database-grounded intervention is evaluated on DialMAGE using the static MultiWOZ database and simulated booking. The objective check depends on the reader's own reading of the pending request.

7 Ethics

EmoWOZ is released under CC BY-NC 4.0 [Feng et al., 2022]; we use it for academic research and do not redistribute the data. All generated replies are from dialogue systems, not real users. No personal information beyond what is in the public corpus was used.

8 Conclusion

Users tell a system it has failed mostly by repeating themselves, and systems answer the repetition instead of the need behind it. Making that need explicit — what was asked, what was delivered, what is still pending, with real data to deliver it — lets a small model do what a strong model does from a prompt, and makes both deliver real, correct entities more often.

Reproducibility. Code, prompts, model outputs and judgments:
`https://github.com/yingchen-coding/frustration-paper`.

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